

blended E27 petrol in Brazil is \$1.15 per litre while E100 costs \$0.77 per litre. In India, most fuel sold is E20, and its retail price is around ₹100 per litre. As blending increases, prices decrease, but in India, E20 is priced the same as E10. The main reason: Fuel prices in India are centrally controlled, and while rising feedstock prices impact ethanol economics, the benefits of blending are yet to be passed on to consumers.

The countries that have got the ethanol economics right have taken different approaches. Brazil has encouraged the use of sugarcane-based ethanol through tax policies, government pricing mechanisms, and widespread availability of flex-fuel vehicles, which let drivers switch between ethanol and gasoline based on price. By keeping ethanol taxes low and supporting local production, it has made ethanol a consistently cheaper alternative at the pump.

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The US relies on corn-based ethanol and supports its production through initiatives such as the Renewable Fuel Standard, subsidies for corn farmers, and funding for ethanol infrastructure. These measures have cut production costs and ensured steady supply. Both countries have managed to pass on the benefits to consumers. They have another advantage—their crop yield is higher compared to India. “In case of damaged and surplus food grain, the yield (ethanol) per tonne of grain is 450 litres, and for maize, it is 380. However, for GMO corn in the US, it is 420-430 litres. There (US and Brazil), yields per acre are much higher compared to India,” says C.K. Jain, President, Grain Ethanol Manufacturers Association. Agriculture scientists in India are working on improving the seed variety for maize with higher starch content and optimal moisture content. Experts say

grain-based ethanol holds immense potential to drive rural development, enhance farmers’ incomes and ensure year-round ethanol production. “It must be actively promoted through supportive policies, assured feedstock supply and fair pricing mechanism,” says Sanjay Ganjoo, Director General, the Indian Federation of Green Energy.

FIXING DEMAND

In India, stakeholders have called for calibrated approaches that provide economic incentives to build consumer confidence and encourage the shift to ethanol-blended fuel. Despite consumer concerns about potential vehicle damage, the Automotive Research Association of India, citing 2016 and 2021 studies, has said that the use of

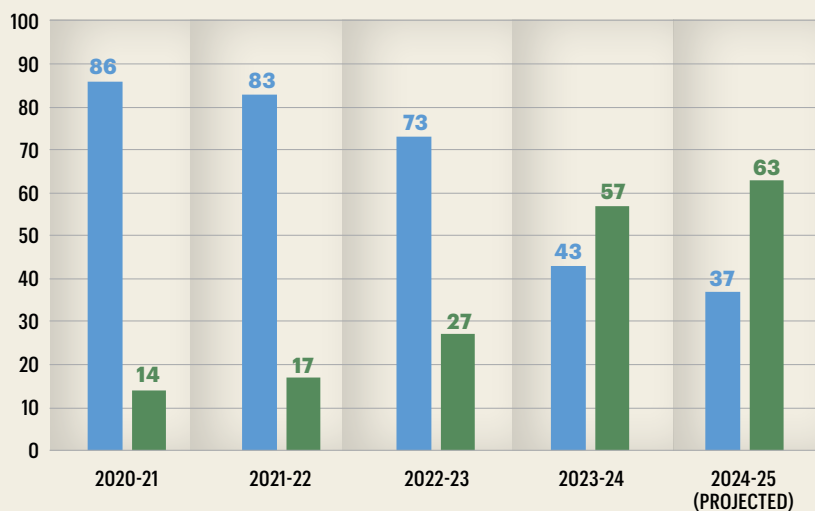
E20 showed no sign of adverse impact on vehicles. Union Minister for Road Transport and Highways Nitin Gadkari accused the petroleum sector of lobbying against the government’s push for ethanol-blended fuel. “Everywhere there are lobbies, there are interests... petrol lobby is very rich,” Gadkari said.

Automakers, while acknowledging a marginal drop in mileage, have reported no complaints of engine damage so far. Experts point out that a 4-6% reduction in fuel efficiency means that consumers will require more fuel to cover the same distance, resulting in increased out-of-pocket spending.

“India had targeted 20% blend for 2030, and we achieved that in 2025. It is a good thing, but the challenge is for the older vehicles. In

SHIFT IN FEEDSTOCK

- Use of sugar for making ethanol is decreasing, as contribution of grains goes up
- Maize’s contribution to grain-based ethanol is around 40%



● SUGAR (%) ● GRAIN (%)

NOTE ETHANOL SUPPLY YEAR CYCLE IS FROM 1ST NOV TO 31 OCT NEXT YEAR

SOURCE NITI AAYOG